ダイナミカルシステムの
過渡応答と安定性について（2）

制御工学１ ⑤

機械理工学専攻
細田　耕

本日の授業のゴール

- ２次系の応答
- 極･零点と過渡応答

２次系（または2次遅れ系）

$$G(s) = \frac{b_0}{s^2+a_1 s+a_2} = \frac{K{\omega_n}^2}{s^2+2\zeta\omega_n s+{\omega_n}^2}$$

システムの極（分母多項式＝0の根）は，

$$s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$$

$0 < \zeta < 1$（虚数根）

$\zeta = 1$（重根）

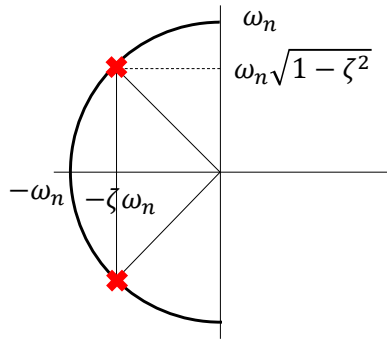
$\zeta > 1$（二つの実数根）

によって，解の性質は異なる

$0 < \zeta < 1$（虚数根）の場合

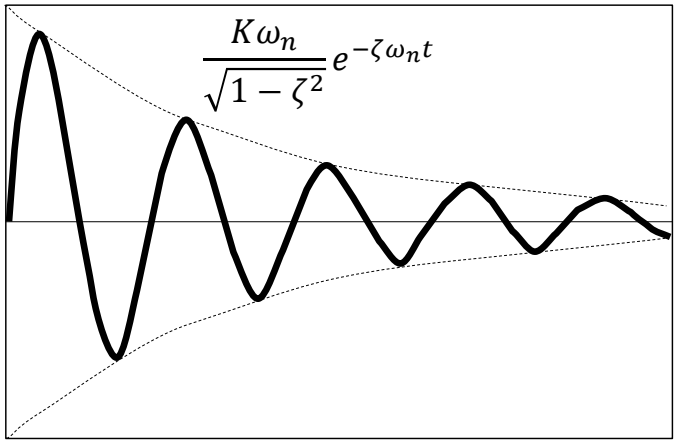
システムの極は，

$$s = -\zeta\omega_n \pm \omega_n\sqrt{1-\zeta^2}j$$
$$= -\zeta\omega_n \pm \omega_d j$$



インパルス応答は，

$$\mathcal{L}^{-1}\left[\frac{K{\omega_n}^2}{s^2+2\zeta\omega_n s+{\omega_n}^2}\right] = \mathcal{L}^{-1}\left[\frac{\sqrt{1-\zeta^2}}{(s+\zeta\omega_n)^2+(1-\zeta^2){\omega_n}^2}\cdot\frac{K{\omega_n}^2}{\sqrt{1-\zeta^2}}\right]$$
$$= \frac{K\omega_n}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin\omega_n\sqrt{1-\zeta^2}t$$
$$= \frac{K\omega_n}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin\omega_d t$$



$0 < \zeta < 1$（虚数根）の場合

ステップ応答は，

$$\mathcal{L}^{-1}\left[\frac{K{\omega_n}^2}{s^2+2\zeta\omega_n s+{\omega_n}^2}\cdot\frac{1}{s}\right]=\mathcal{L}^{-1}\left[\frac{K}{s}-\frac{K(s+2\zeta\omega_n)}{s^2+2\zeta\omega_n s+{\omega_n}^2}\right]$$

$$=\mathcal{L}^{-1}\left[\frac{K}{s}-\frac{K(s+\zeta\omega_n)}{(s+\zeta\omega_n)^2+{\omega_d}^2}-\frac{K\zeta}{\sqrt{1-\zeta^2}}\cdot\frac{\sqrt{1-\zeta^2}\omega_n}{(s+\zeta\omega_n)^2+{\omega_d}^2}\right]$$

$$=K\left\{1-e^{-\zeta\omega_n t}\left(\cos\omega_d t+\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right)\right\}$$

$$=K\left\{1-\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t+\theta)\right\}$$

ここで

$$\theta=\tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}$$

$0 < \zeta < 1$（虚数根）の場合

ステップ応答は，

$$\mathcal{L}^{-1}\left[\frac{K{\omega_n}^2}{s^2+2\zeta\omega_n s+{\omega_n}^2}\cdot\frac{1}{s}\right]=\mathcal{L}^{-1}\left[\frac{K}{s}-\frac{K(s+2\zeta\omega_n)}{s^2+2\zeta\omega_n s+{\omega_n}^2}\right]$$

$$=\mathcal{L}^{-1}\left[\frac{K}{s}-\frac{K(s+\boxed{\zeta\omega_n})}{(s+\zeta\omega_n)^2+{\omega_d}^2}-\frac{K\zeta}{\boxed{\sqrt{1-\zeta^2}}}\cdot\frac{\boxed{\sqrt{1-\zeta^2}}\omega_n}{(s+\zeta\omega_n)^2+{\omega_d}^2}\right]$$

$$=K\left\{1-e^{-\zeta\omega_n t}\left(\cos\omega_d t+\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right)\right\}$$

$$=K\left\{1-\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t+\theta)\right\}$$

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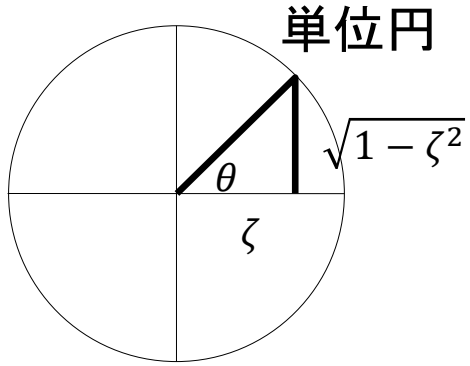


$\zeta = 1$（重根），$\zeta > 1$（実数根）の場合

$\zeta = 1$（重根）の場合のステップ応答は

$$K\{1 - e^{-\omega_n t}(1 + \omega_n t)\}$$

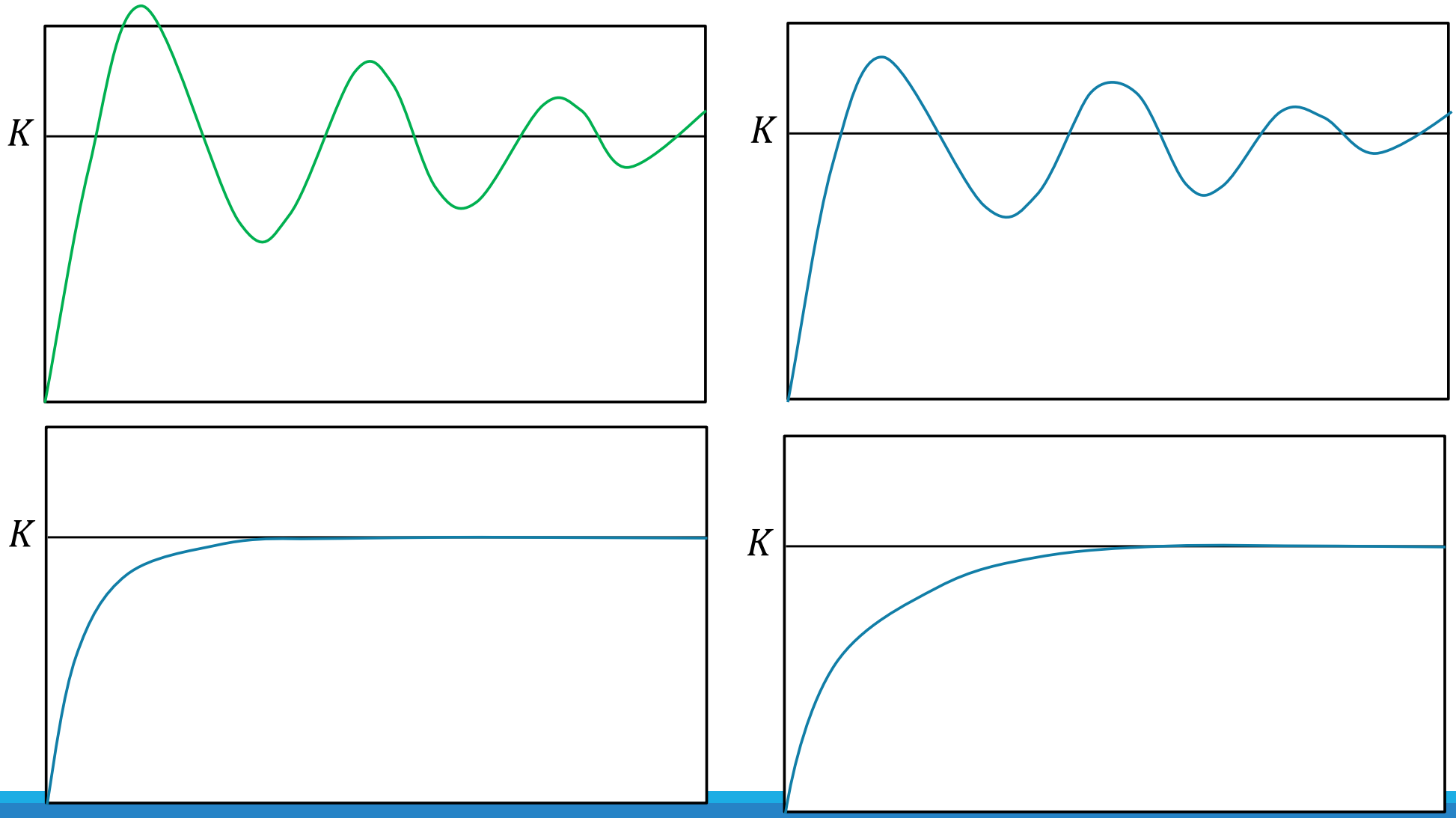
$\zeta > 1$（実数根）の場合のステップ応答は

$$K\left\{1 - \frac{e^{-\zeta\omega_n t}}{2\beta}\left((\zeta + \beta)e^{\omega_n \beta t} - (\zeta - \beta)e^{-\omega_n \beta t}\right)\right\}$$

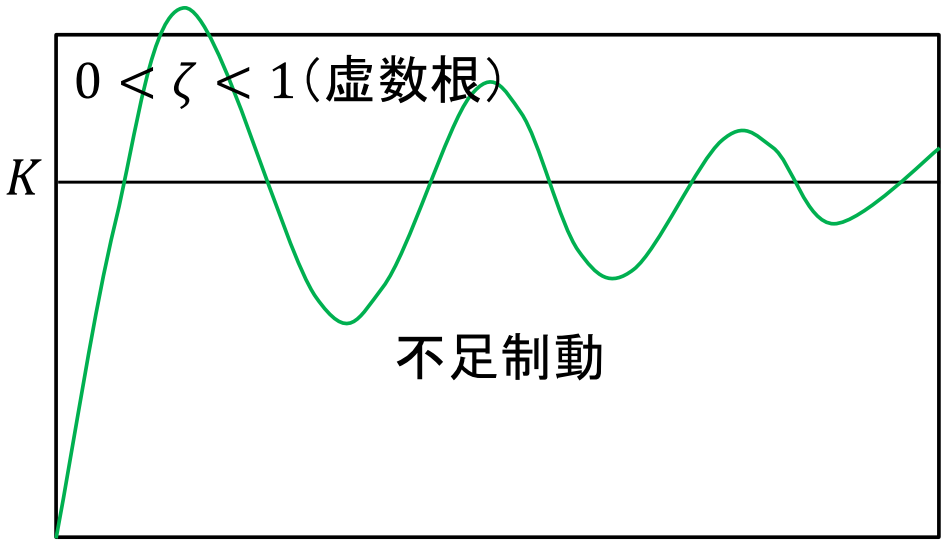
ただし

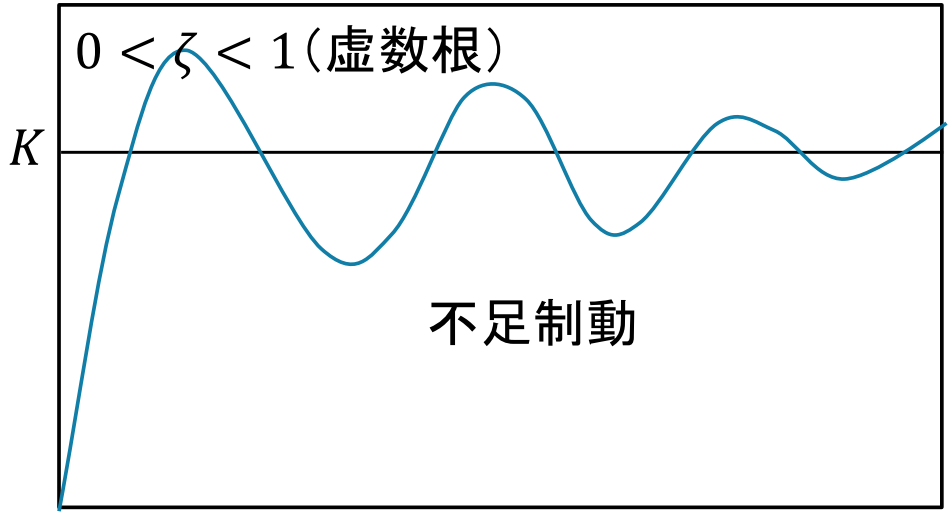
$$\beta = \sqrt{\zeta^2 - 1}$$

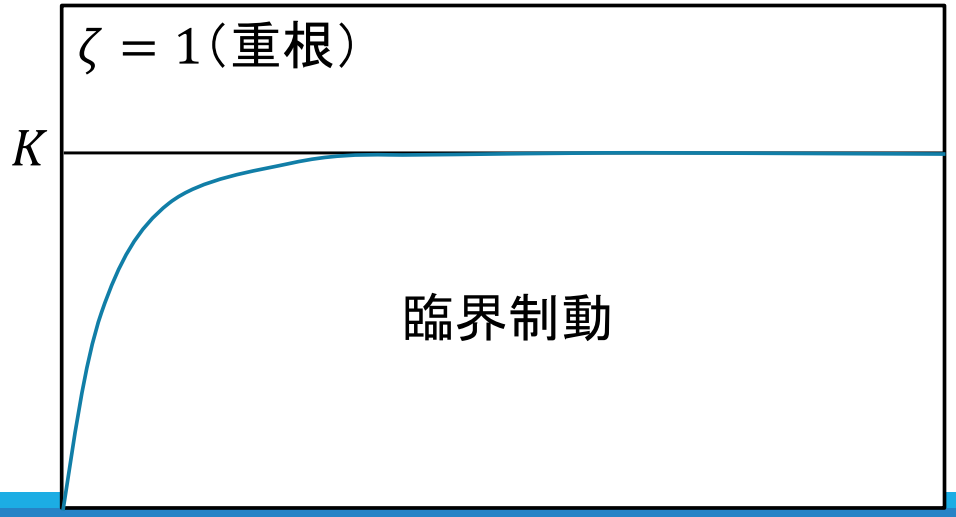
2次系のステップ応答（まとめ）

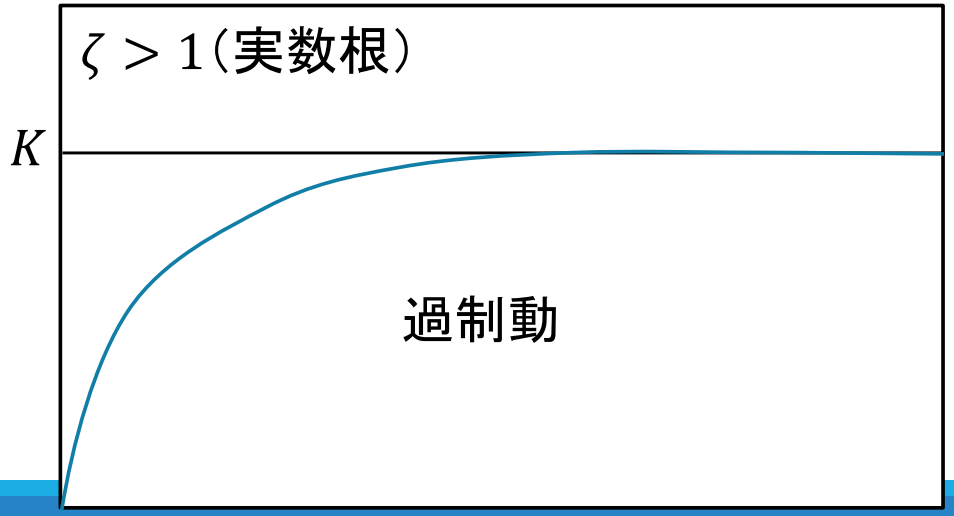


2次系のステップ応答（まとめ）



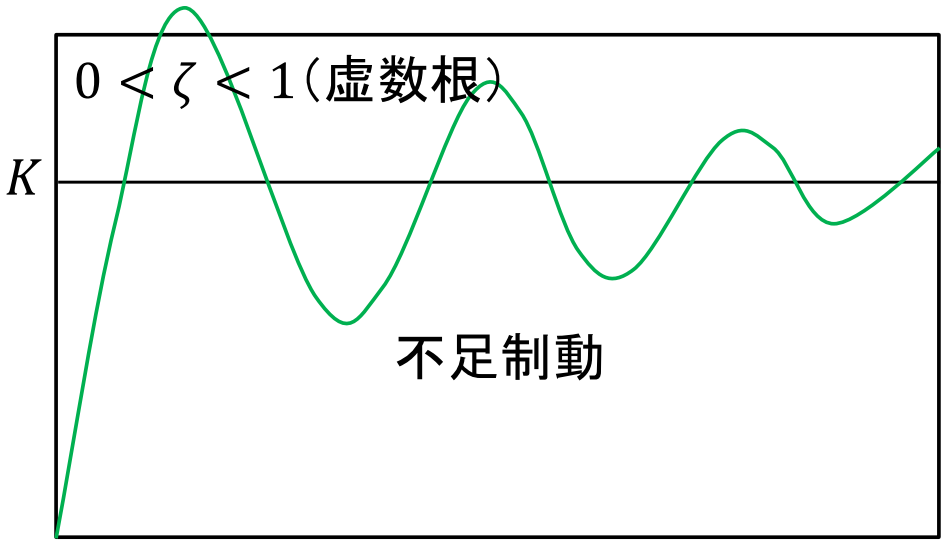


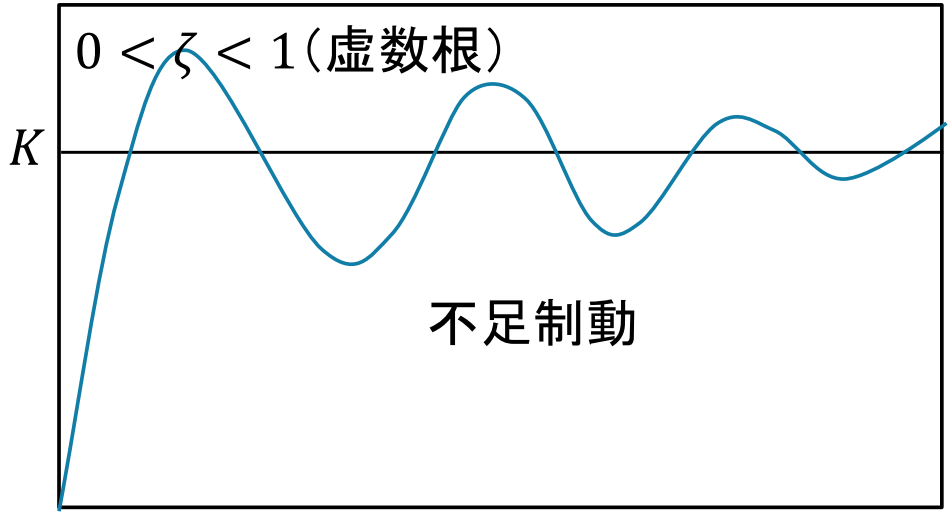


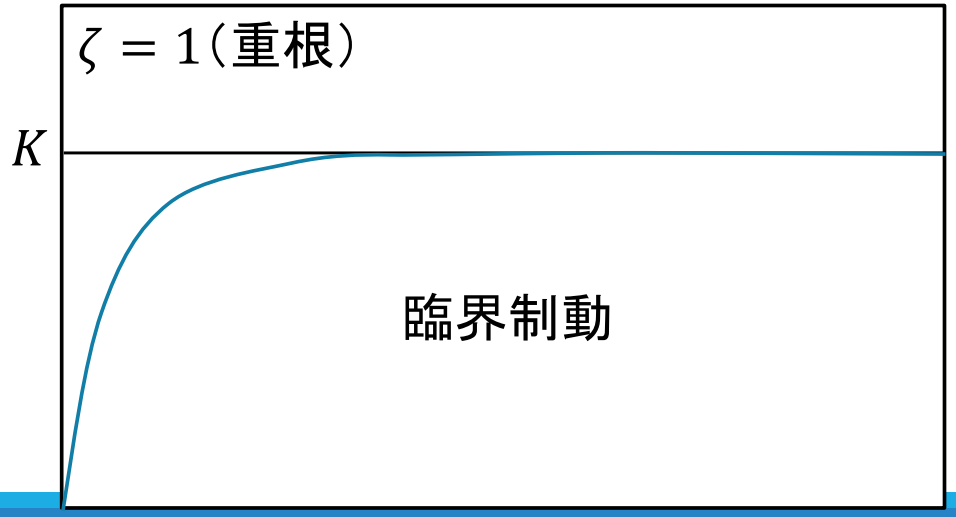


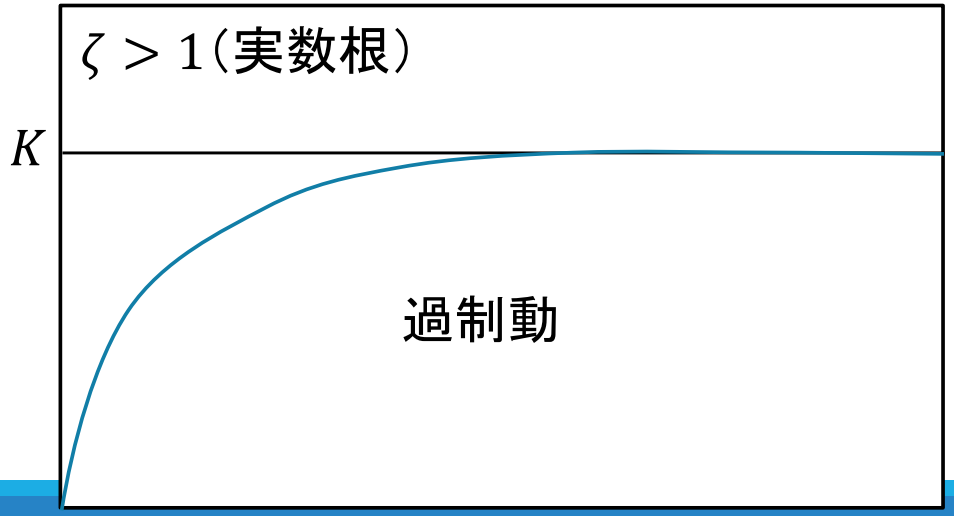
2次系のステップ応答（まとめ）

ζ : 減衰係数
ω_n : 自然角周波数
K : ゲイン

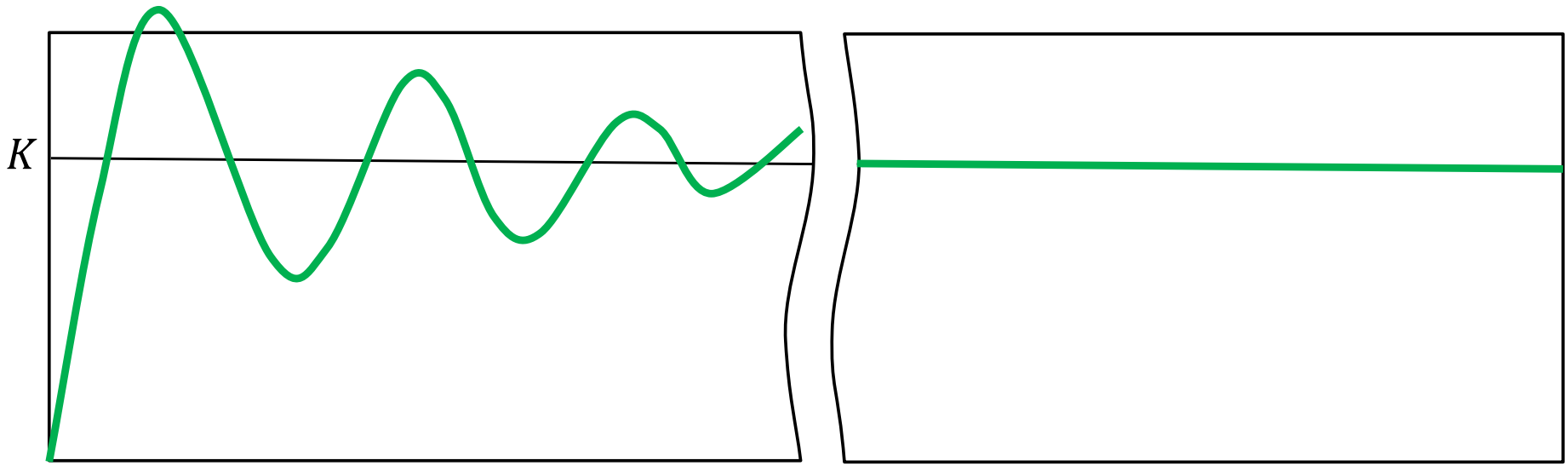








過渡応答



極・零点と過渡応答

ダイナミカルシステム

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0} \qquad (n \geq m)$$

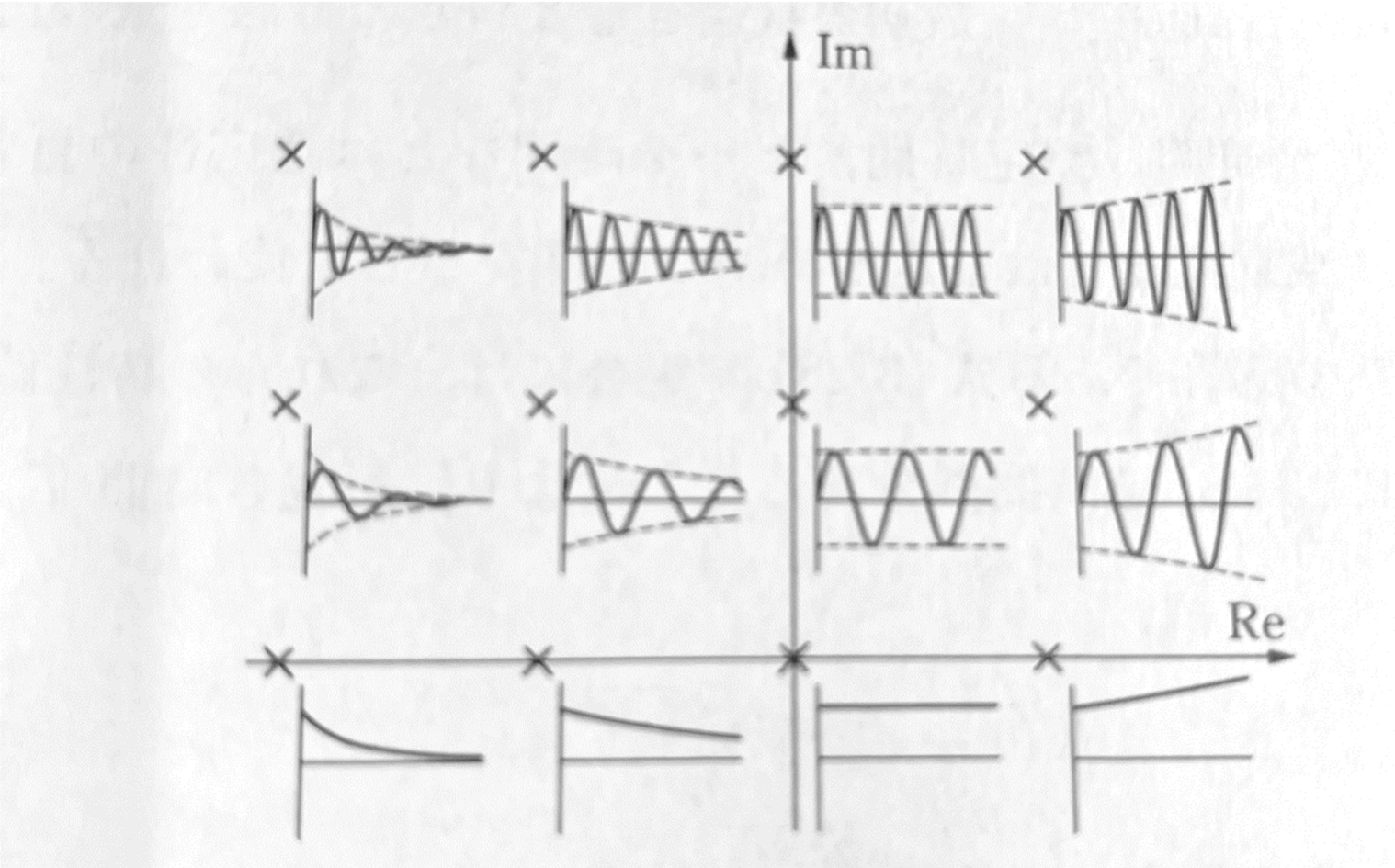
分母多項式=0の解が$\sigma_1, \sigma_2, \cdots, \sigma_n$であるとすると，ステップ応答は，

$$\mathcal{L}^{-1}\left[G(s)\frac{1}{s}\right] = \mathcal{L}^{-1}\left[\frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{(s-\sigma_1)(s-\sigma_2)\cdots(s-\sigma_2)}\right]$$

$$= \mathcal{L}^{-1}\left[\frac{k_1}{s-\sigma_1} + \frac{k_2}{s-\sigma_2} + \cdots + \frac{k_n}{s-\sigma_n} + \frac{k_0}{s}\right]$$

$$= k_0 + k_1 e^{\sigma_1 t} + k_2 e^{\sigma_2 t} + \cdots$$

極・零点と過渡応答



本日の授業のゴール

- ２次系の応答（2次系の応答は重要な割に試験には出ない）
- 極・零点と過渡応答